



Software Architecture 2025-2026

Lecture 1, part 2/2

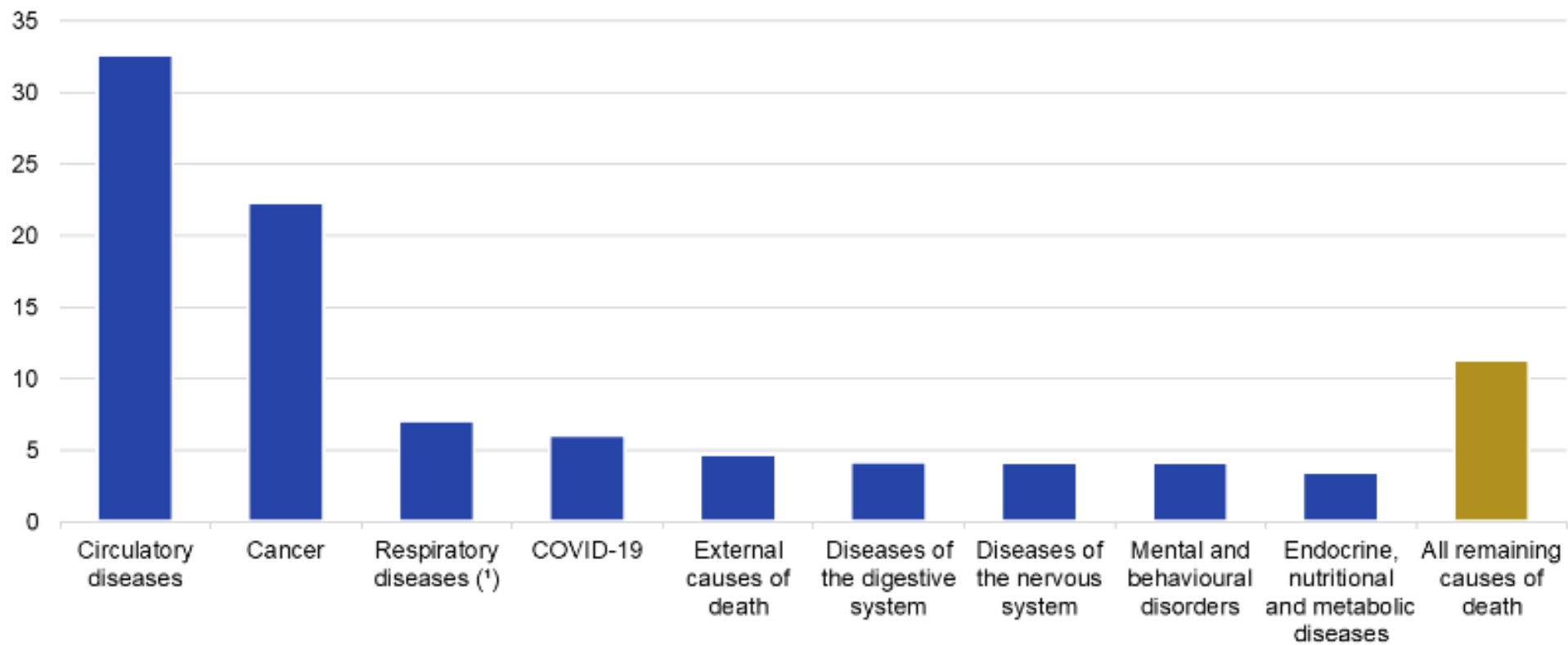
Introduction to the application case

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Share of main causes of death, EU, 2022

(%)



⁽¹⁾ Excluding COVID-19.

Source: Eurostat (online data code: hlth_cd_aro)

Main causes of death by age and sex, EU, 2022

(standardised death rates per 100 000 residents)

	Men		Women	
	Cause of death	Rate	Cause of death	Rate
Aged less than 65 years	Ischaemic heart disease	27.1	Breast cancer	12.0
	Accidents	21.4	Lung cancer (*)	9.9
	Lung cancer (*)	18.0	Ischaemic heart disease	6.2
	Unknown and unspecified causes	17.0	Unknown and unspecified causes	5.4
	Chronic liver disease (²)	14.5	Accidents	5.1
	Intentional self-harm	13.8	Colorectal cancer (*)	5.1
Aged 65 years and over	Ischaemic heart disease	654.2	Ischaemic heart disease	382.0
	COVID-19 (³)	405.5	Cerebrovascular diseases	301.6
	Cerebrovascular diseases	357.2	COVID-19 (³)	221.9
	Lung cancer (*)	278.9	Dementia	181.6
	Chronic lower respiratory diseases	182.6	Lung cancer (*)	113.1
	Prostate cancer	171.6	Breast cancer	106.4

(¹) Malignant neoplasms of the trachea, bronchus and lung.

(²) Cirrhosis, fibrosis and chronic hepatitis

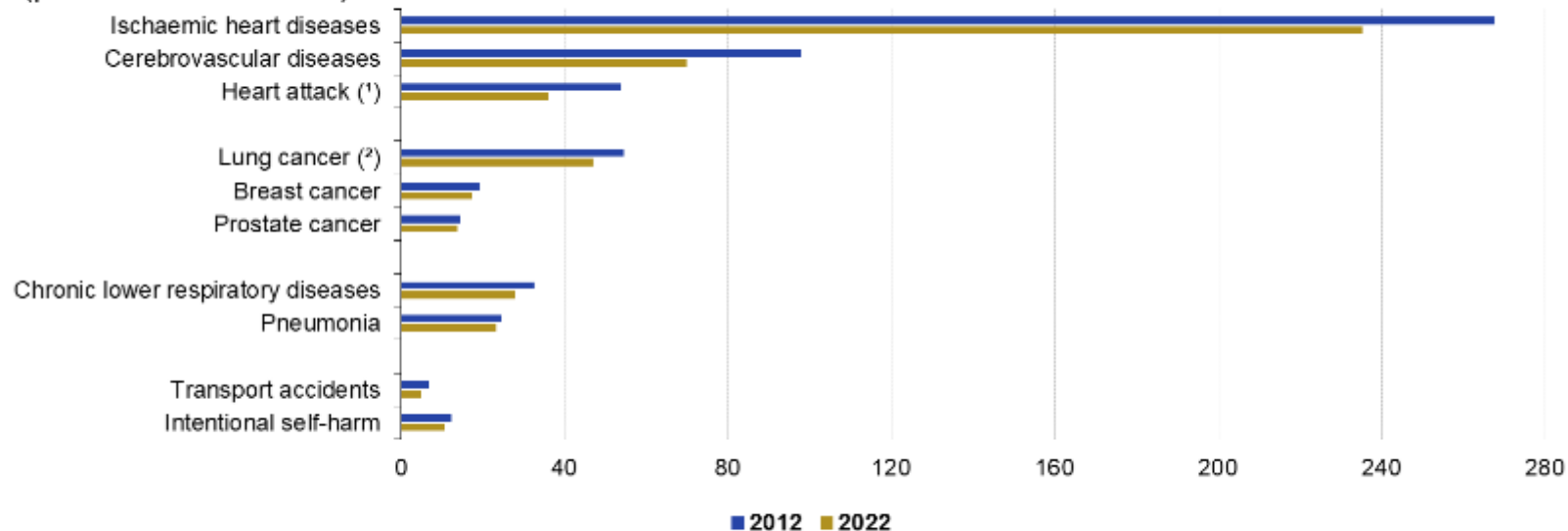
(³) See the section on the classification of the causes of death for details about how statistics on COVID-19 deaths were calculated.

(*) Malignant neoplasm of colon, rectum and anus

Source: Eurostat (online data code: hlth_cd_asdr2)

Standardised death rates from selected causes of death, EU, 2012 and 2022

(per 100 000 residents)



⁽¹⁾ Acute myocardial infarction including subsequent myocardial infarction

⁽²⁾ Malignant neoplasms of the trachea, bronchus and lung.

Source: Eurostat (online data code: hlth_cd_asdr2)

eurostat 

Application case

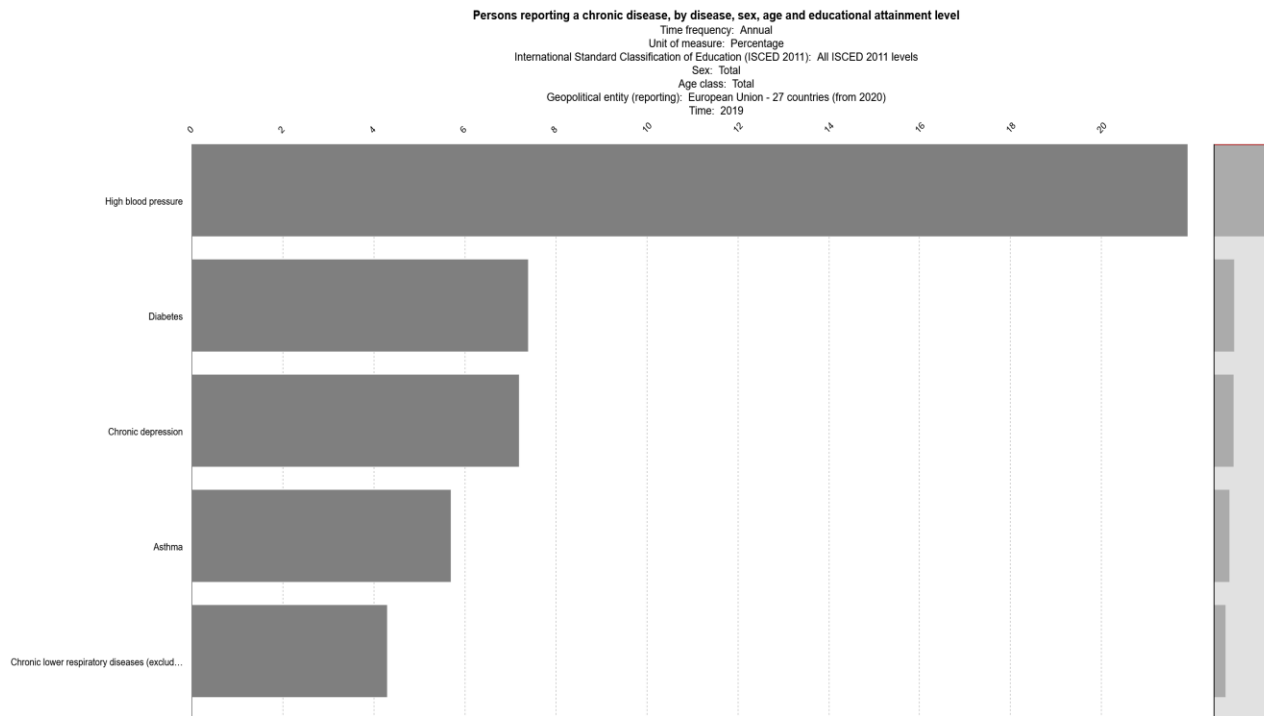
PMS: a Patient Monitoring Service for cardiovascular disease (CVD)

Cardiovascular diseases (CVD)

- › a broad group of medical problems that affect the **circulatory system** (the heart and blood vessels),
 - ›› often resulting from **atherosclerosis**, the abnormal build-up of plaque in the patient's arteries
- › Leads to **ischaemic heart disease** (heart attacks) and **cerebrovascular diseases** (strokes)

Context

- › Persistent effect of **high blood pressure** in arteries may lead to **chronic failure of vital organs**
- › **Chronic disease**, slowly evolving
- › Lifestyle, genetics, ..



Persons reporting a chronic disease, by disease, sex, age and educational attainment level [hlth_ehis_cd1e\$defaultview]

Source of data: Eurostat - Last updated date: 03/01/2024 23:00

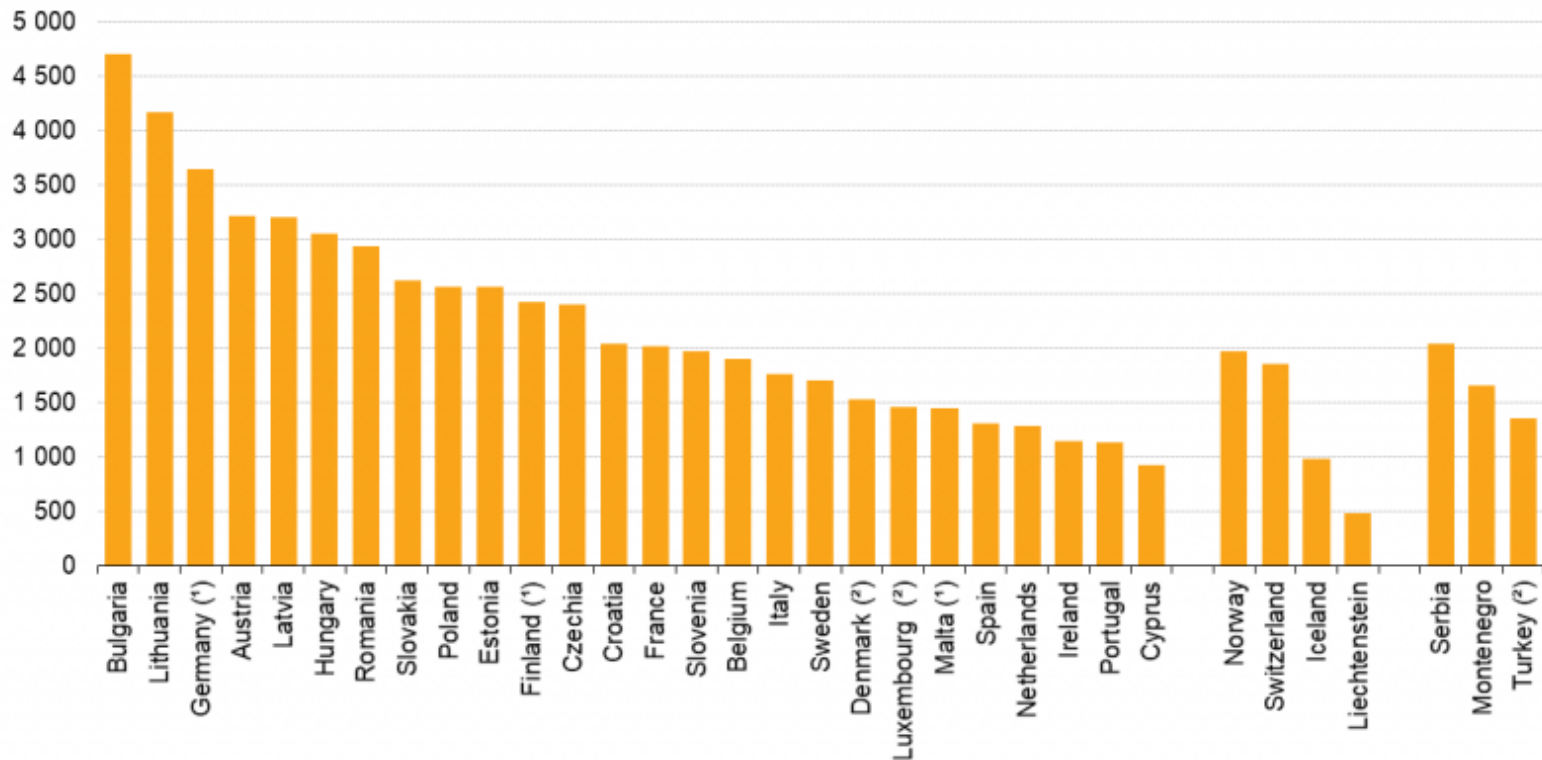
Disclaimer This graph has been created automatically by ESTAT/EC software according to external user specifications for which ESTAT/EC is not responsible. Graphic included.

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Context

CVD is a cause of **pressure** on hospitals and healthcare systems

Hospital discharge rates for in-patients with diseases of the circulatory system, 2019
(per 100 000 inhabitants)



Note: Greece, data not available.

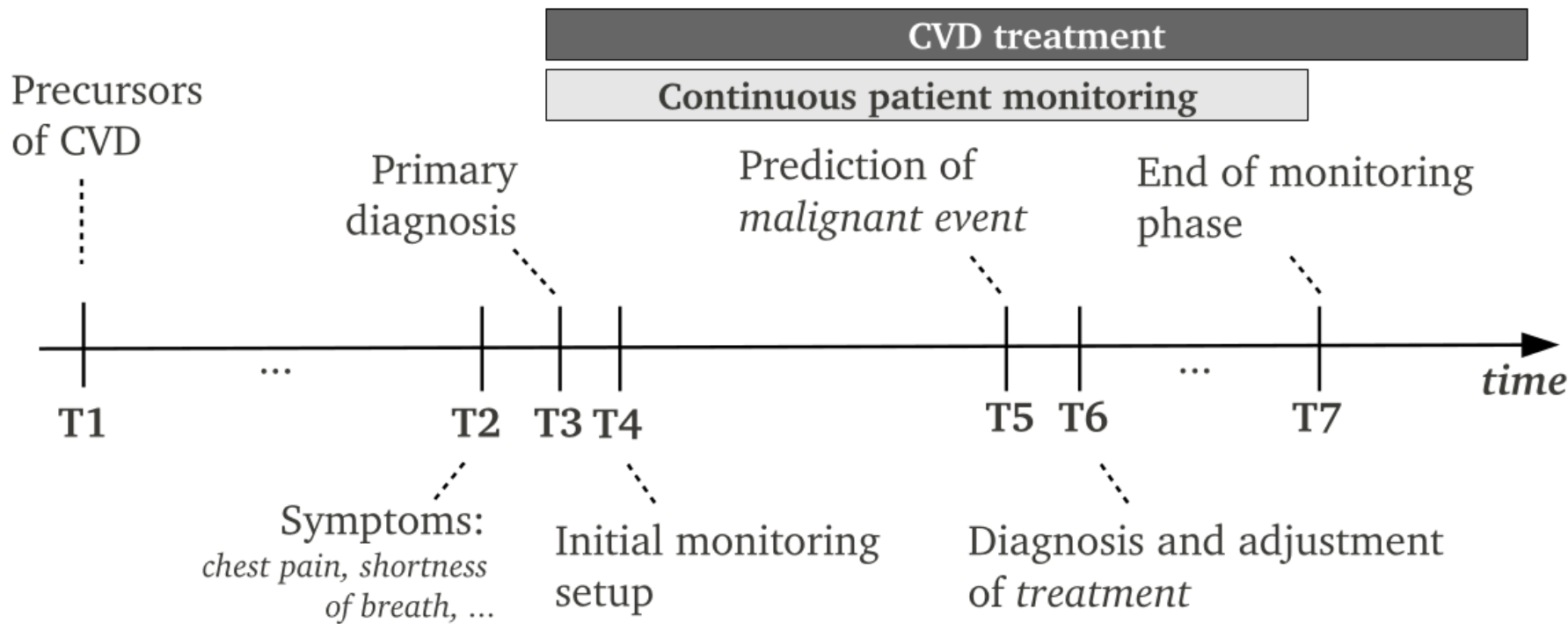
⁽¹⁾ 2018.

⁽²⁾ 2016.

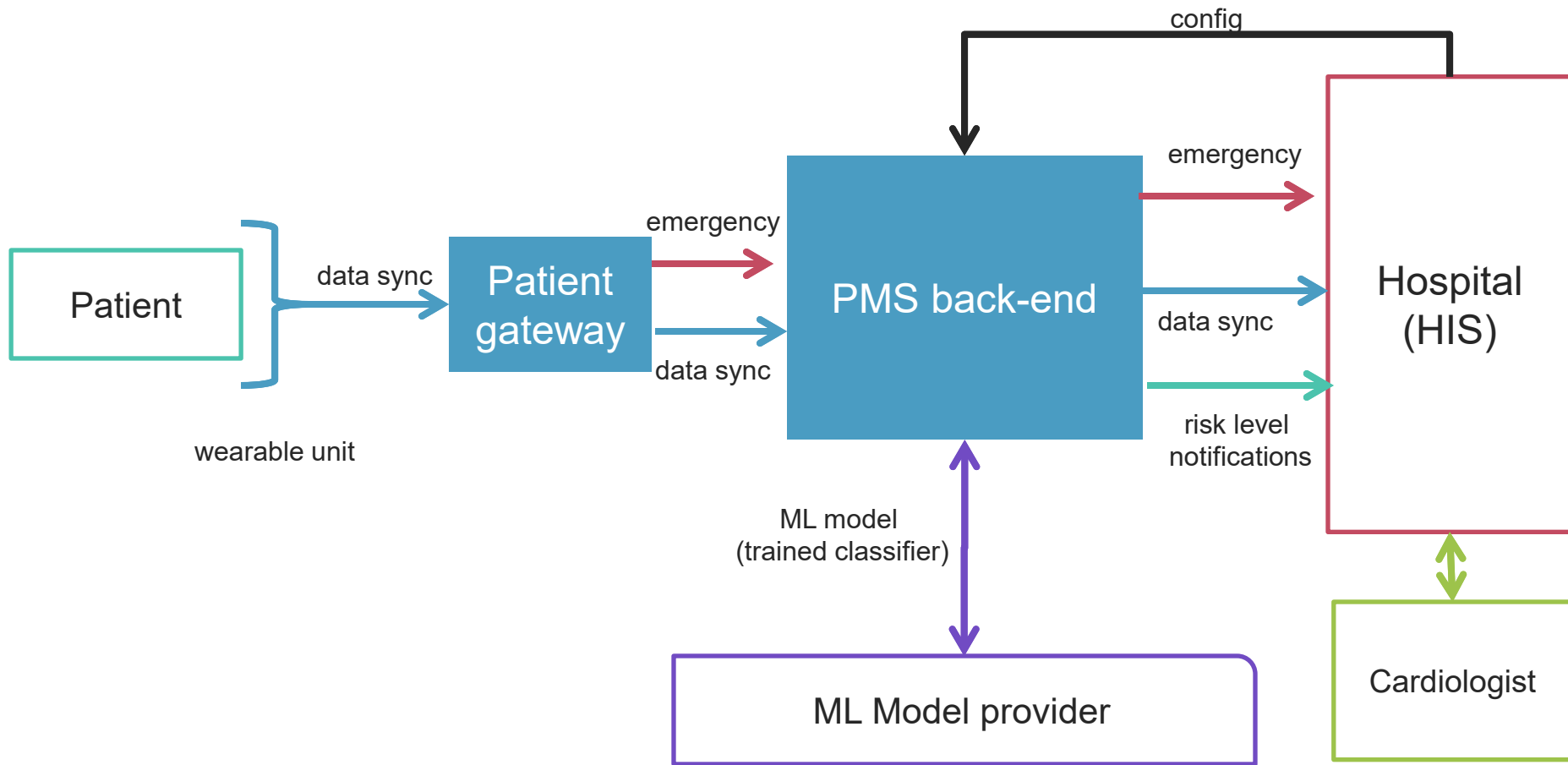
Source: Eurostat (online data code: hlth_co_disch2)

Positioning

- › Potential in longer-term management and follow-up of the evolution of CVD, outside of hospital context (“extra-muros”)
- › Telemedicine, data centric, and IoT-based service:
 - › Remote and continuous monitoring / data collection
 - › Prediction of malignant events at the basis of collected data
 - › Timely decision making
 - › Limited impact on patient’s day to day life
- › We are a small startup company, with strong ties to one specific hospital, building a pilot system



Overview and positioning



Wearable unit

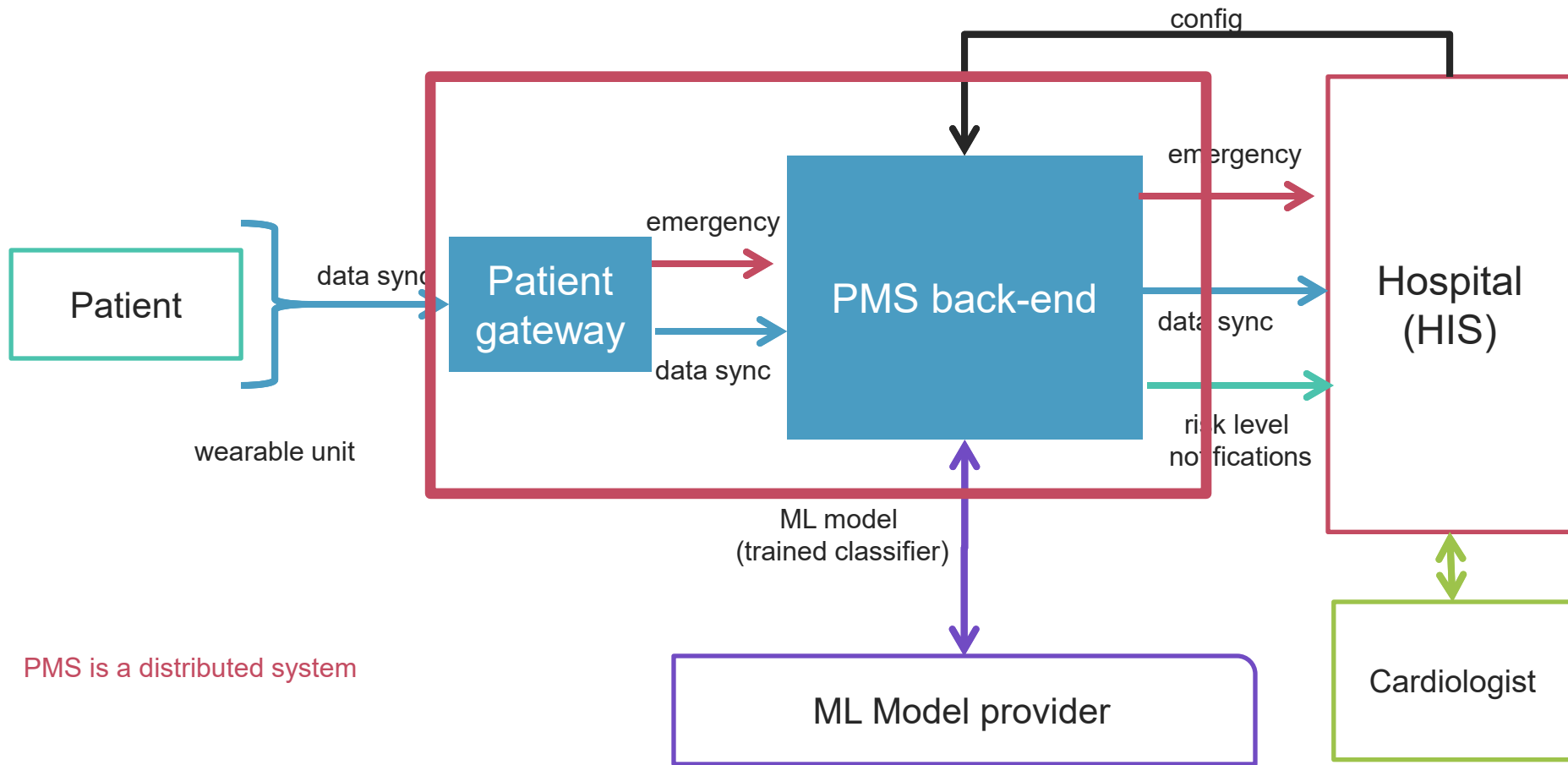
- › a **medical device**, fitted with a collection of sensors capable of acquiring relevant parameters
 - › Heart rate
 - › Electrocardiograph (ECG)
 - › Respiratory rate
 - › Blood oxygen level
 - › Blood pressure
 - › Activity level (accelerometer)
 - › Body temperature
- › Battery-operated, bluetooth connectivity



Patient gateway

- › = a smartphone with a dedicated app
- › Relays data to the back-end at fixed intervals
 - ›› Interval depends on patient status/configuration
 - ›› On-demand consultation, upon request
- › Performs emergency assessment
- › UI for patient

Overview and positioning



Automated risk estimation at the heart

Clinical risk models

- › **Inputs:** new data, historical data, patient pathologies, current risk level, (persisted artifacts such as config, trained classifier)
- › Output: suggest a **risk level (low, medium, high)**
 - › Notifications can be sent in consequence to treating physician (or emergency services)

No single model suits all pathologies/patients

- › Configured on a per-patient basis (combination, thresholds, etc)
- › A single risk estimation=weighted combination of the outcome of individual models

Automated risk estimation at the heart

Different levels of complexity

Risk level	Normal	Worrisome	Dangerous
Low	105–120	90–105, 120–150	80–90, 150–190
Medium	95–135	90–95, 135–160	80–90, 160–190
High	90–150	85–90, 150–170	80–85, 170–190

Table 2: Default thresholds for systolic blood pressure (mmHg).

Notice the wider ranges for patients with higher risk-levels. Because of their cardiovascular conditions, their normal values will be, for example, more elevated, than the normal values of low-risk patients.

Automated risk estimation at the heart

Different levels of complexity

- › Machine learning approach that involves pre-trained classifiers
 - › External party = **CVD research institute** that optimizes different ML models for different pathologies
 - › We will download these and use them
 - › All **risk estimation** at the basis of **actual patient data** is performed by us
- › Lightweight risk estimation for **quick and reliable detection of emergencies** on the patient gateway

Stakeholders

- › Patients
- › Physicians (GP, cardiologist)
- › Nurse (trained nurse for registration)
- › Emergency call center/services
- › Telemedicine operators
- › Hospital
- › Electronic Health Record (EHR) services
- › Legal/financial department
- › Telecom operator
- › ML model provider

Constraints and considerations

- › Hospital information system, and applicable HL7 FHIR (Fast Healthcare Interoperability Resources) standard
 - ›› Physician workstation: request on-demand consultation, follow-up on evolution, change monitoring configuration
 - ›› Patient enrollment, by a trained nurse
 - ››› associating a Wearable unit to a patient, linking to the patient record, initializing the risk estimation, correct association to the treating physician
- › Security/privacy/legal
- › Financial

PMS Functional requirements

... in Appendix A

It's all about the NFRs

